MILITARY SPECIFICATION

TEST SET, RADAR
AN/APM-299

This Specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1.0 SCOPE

1.1 Scope - The Test Set, Radar AN/APM-299, hereinafter referred to as Test Set, covered by this specification is for Intermediate level use and shall facilitate Radar Systems alignment and testing.

1.2 Classification - The Test Set shall consist of the following items:

Item	Type Designation	Appl. Para.
Test Set, Subassembly	TS-2797/APM-299	3.4.1
Case, Test Set	CY-6615/APM-299	3.4.2
Cover, Test Set	CW-1048/APM-299	3.4.3
Cable Assembly, Power (W1)		3.4.1.3(p)(2)
Cable Assembly, Delayed Trigger Input (W2)		3.4.1.3(p)(3)
Cable Assembly, Sync Input (W3)		3.4.1.3(p)(4)
Cable Assembly, COHO Frequency Check (W4)		3.4.1.3(p)(5)
Cable Assembly, COHO Input (W5)		3.4.1.3(p)(6)
Cable Assembly, 216 Mc Output (W6)		3.4.1.3(p)(7)
Cable Assembly, Doppler Input (W7)		3.4.1.3(p)(8)

FSC 6625

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2.0 APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedure; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-S-23603	System Readiness/Maintainability, Avionic Systems Design, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Experimental Distribution

DRAWINGS

Military

MS17322	Meter, Time Totalizing 115 Volt, 400-Cycle, Rev. B
MS21993	Cable Assembly, Power, Electrical
MS3107	Connector, Plug Electric, Quick Disconnect

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2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3.0 REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Microelectronics items shall conform to requirements specified herein.

(b) Other parts and materials requirements shall conform to specification MIL-T-21200.

(c) Nonrepairable subassemblies, as outlined in MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

(d) When previously produced models of this Test Set did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall be described by a contractor procurement specification and must be approved by the procuring activity. Microelectronic Modular Assemblies (Microelectronic Circuits) (MICS) shall be contained within Maintenance Modules (for example, on printed circuit cards).

3.1.2 Modules, Maintenance - The electronic portions of the Test Set shall be divided into maintenance modules (see MIL-S-23603). Maintenance modules shall normally be considered repairable.

3.2 Design and Construction - The Test Set shall conform with all the applicable requirements of MIL-T-21200 for design, construction and workmanship except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the Test Set, including cables, shall be a minimum consistent with good design and shall not exceed 135 pounds.

3.2.2 Reliability

3.2.2.1 Operational Stability - The Test Set shall operate with satisfactory performance, continuously or intermittently, for a period of at least 24 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operation Life - The Test Set shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

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3.2.2.3 Reliability in Mean Time Between Failure (MTBF) - The Test Set shall be tested in accordance with the requirements of MIL-STD-781, paragraph 4.1.1, for a period of 12 hours. The test results shall indicate a MTBF of not less than two hundred (200) hours.

3.2.2.4 Time Totalizing Meter - The Test Set shall contain time totalizing meters in accordance with MIL-M-7793. Meter shall be included in the following unit:

Unit	Type of Meter
Test Set, Radar TS-2797/APM-299	MS 17322

3.2.3 Cables and Connectors - The Test Set shall provide for the use of cables and connectors in accordance with MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.2.3.1 Cable Assemblies, Power Electrical - Applicable input power cable and connections shall, in general, conform to MS21993 with the exception that the cable shall be 6 feet in length.

3.2.4 Interchangeability - The Test Set shall meet the interchangeability requirements as defined in MIL-T-21200.

3.2.5 Interference Control - The generation of radio interference by the Test Set and its vulnerability to radio interference shall be controlled within the limits of MIL-I-6181.

3.2.6 Maintenance, Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approved for equipment identification shall be in accordance with MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° $\pm 10^{o}$C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 VAC 400 ±5 cps, 3ϕ

3.2.9 Service Conditions - The Test Set shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for class 3 equipment except as modified herein.

3.2.10 Warm-up Time - The time required for the Test Set to warm-up prior to operation shall be kept to a minimum and shall be indicated by illumination of the front panel OPERATE lamp. The warm-up time shall not exceed 25 minutes at extreme service conditions and should be approximately 15 minutes under standard conditions.

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3.2.11 Primary Input Power Requirements - The Test Set shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-21200. The power required shall be of the following type(s) and shall not exceed the specified amount.

AC Power, Three Phase, 115/200 V, 380-420 cps, 350 VA

3.2.11.1 Protection - The Test Set shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

3.3 Performance - The performance of the Test Set shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.3.1 Outputs

(a) Two X-Band Pulses, coincident in time, separated in frequency by 216 Mc. This output shall be available at the test set front panel with a UG 136B/U waveguide flange.

(b) One 216 Mc Pulse for COHO lock. This output available at front panel by means of a female BNC connector.

(c) COHO Frequency check signal or Recovered Doppler signal available at test set front panel by means of a female BNC connector.

(d) X-Band Delta Frequency, available at test set front panel by means of a female BNC connector.

(e) 216 Mc CW Output, available at front panel of test set when Meter Selector switch is in 216 Mc power calibrate position only. This output can be used for 216 Mc frequency counter check. Available from same output connector as indicated in (b).

3.3.2 Inputs

(a) Doppler Frequency, front panel female BNC connector.

(b) COHO Frequency, front panel female BNC connector.

(c) Sync Pulse, front panel female BNC connector.

(d) Delayed Trigger Pulse, front panel female BNC connector.

(e) Prime Power, 400 cycles, 3 ϕ, Y connection, 115 VAC Line-to-Neutral. Connector to be MS-3107A-20-4P.

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3.4 Detail Requirements

3.4.1 Test Set, Subassembly TS-2797/APM-299 - The Test Set subassembly TS-2797/APM-299, housed in the Case, Test Set CY-6615/APM-299, is a Pulse Doppler X-Band Signal Generator. Through external triggering and audio modulation, two X-Band pulses are generated simultaneously at the output, which has the simulated doppler modulation present for use as test signals to facilitate testing and alignment of the Radar System.

3.4.1.1 Form Factor - The Test Set subassembly TS-2797/APM-299 shall be housed in the Case, Test Set CY-6615/APM-299 which is a combination case and fulfills the requirements of MIL-T-21200. The overall dimension of the Test Set with cover installed shall not exceed 23 inches by 33 inches by 18 inches.

3.4.1.2 Weight - The maximum weight of Test Set, Radar AN/APM-299 shall be 135 pounds.

3.4.1.3 Content - The Test Set Subassembly shall contain the following major assemblies, subassemblies, and accessory items:

(a) X-Band Source - The X-Band Source shall be an internal cavity klystron capable of tuning over the range of 8.5 Gc to 9.4 Gc. It shall be operated in the CW mode only.

(b) X-Band Air Strip - The Air Strip shall accept the X-Band energy and shall be used primarily for signal routing purposes.

(c) X-Band Single Sideband Modulator - The X-Band Single Sideband Modulator shall accept the X-Band Source energy routed through the Air Strip. It shall also accept a doppler modulated 216 Mc signal and shall produce an output signal consisting of the X-Band Source frequency shifted by 216 Mc plus the doppler modulation.

(d) X-Band Tunable Filter - The X-Band Tunable Filter shall be mechanically tracked with the X-Band Source and shall pass only the X-Band Fo + 216 Mc signal from the X-Band Single Sideband Modulator.

(e) X-Band Air Strip - The Air Strip shall accept both the X-Band Source frequency from the Air Strip and the Fo + 216 Mc signal from the X-Band Tunable Filter. It shall combine both signals for application to the X-Band Switch. In addition, the Air Strip will provide power monitoring outputs for both the Fo and the Fo + 216 Mc signals.

(f) X-Band Switch - The X-Band Switch shall accept both the Fo and Fo + 216 Mc CW signals from the Air Strip and shall cause both frequencies to be pulse modulated.

(g) X-Band Attenuator - The X-Band Attenuator shall attenuate both the Fo + 216 Mc signals over a 90 db range.

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(h) X-Band Isolator - The X-Band Isolator shall prevent load VSWR or return power from affecting the operation of the Test Set. The output of this isolator shall form the output flange of the Test Set.

(i) X-Band Thermistor Mount - The X-Band Thermistor Mount shall be mounted directly on the Air Strip and shall form a part of a thermistor bridge for monitoring Fo power. It shall be used for output power calibration of the Test Set.

(j) 216 Mc Source - The 216 Mc Source shall generate a highly precision 216 Mc CW frequency for use as the basic Test Set reference.

(k) RF Component Mounting Subassembly To Consist Of The Following:

(1) 216 Mc Hybrids - The 216 Mc Hybrids mounted on RF Component Mounting Subassembly shall be primarily for 216 Mc power splitting for signal routing purposes.

(2) 216 Mc Fixed Attenuators - The 216 Mc Fixed Attenuators mounted on the RF Component Mounting Subassembly shall be used for power level adjustment in the signal routing path.

(3) 216 Mc Amplifiers - The 216 Mc Amplifiers shall be used for amplification and adjustment of 216 Mc CW power to the 216 Mc switch and shall be used for amplification and adjustment of the doppler modulated 216 Mc CW power to the X-Band Single Sideband Modulator.

(4) 216 Mc Switch - The 216 Mc Switch shall accept the 216 Mc CW signal from the 216 Mc Amplifier and shall cause the 216 Mc frequency to be pulse modulated.

(5) 216 Mc Power Monitor - The 216 Mc Power Monitor shall be used for monitoring the 216 Mc power. It shall be used for 216 Mc output power calibration of the Test Set.

(6) 216 Mc Mixer - The 216 Mc Mixer shall accept an input from the 216 Mc Source and an input from the radar COHO. It shall mix these two frequencies and shall produce a difference frequency for application to the COHO Freq. Chk Ext. connector.

(l) 216 Mc SSB Modulator - The 216 Mc SSB Modulator shall accept the quadrature doppler audio signals and the 216 Mc source energy. It shall produce an output which consists of the 216 Mc signal shifted by the doppler modulation.

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(m) Power Supply - The Power Supply shall convert the 115V/200V 400 cps input power to the following regulated DC voltages:

(1) + 6 VDC

(2) - 6 VDC

(3) +12 VDC

(4) -12 VDC

(5) +6.3 VDC

(6) -165 VDC

(7) -400 VDC

(8) +28 VDC (Unregulated)

(9) + 5 VDC

(n) Quadrature Network - The quadrature network shall accept the doppler input signal and shall generate two signals at the same frequency as the input but in phase quadrature. These two output signals shall be applied to the 216 Mc Single Sideband Modulator.

(o) Pulse Generator and Driver - The pulse generators shall accept a trigger pulse from the radar and shall regenerate the pulse to a particular pulse width. This regenerated pulse is to be applied to the RF switch to produce pulse modulated RF from the CW signal.

(p) Accessory Items - The following items are to be stored in cover, Test Set, CW-1048/APM-299.

(1) Extender Card - The Extender Card shall be used during maintenance of the Test Set.

(2) Power Cable Assembly (W1) - The Power Cable Assembly (W1) shall consist of the following:

Connector, MS3107-20-4S

Connector, MS3107A-20-4P

Cable, CO-04MGF (4/1S)0360, 6 feet

(3) Delayed Trigger Input Cable Assembly (W2) - The Delayed Trigger Input Cable Assembly (W2) shall consist of the following:

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Connector, UG-1033/U

Connector, UG-1033/U

Cable, RG122/U, 50 ohm, 6 feet

(4) Sync Input Cable Assembly (W3) - The Sync Input Cable Assembly (W3) shall consist of the following:

Connector, UG-1033/U

Connector, UG-1033/U

Cable, RG 122/U, 50 ohm, 6 feet

(5) COHO Freq. Chk Cable Assembly (W4) - The COHO Freq. Chk Cable Assembly (W4) shall consist of the following:

Connector, UG-1033/U

Connector, UG-1033/U

Cable, RG 122/U, 50 ohm, 6 feet

(6) COHO Input Cable Assembly (W5) - The COHO Input Cable Assembly (W5) shall consist of the following:

Connector, UG 250/U

Connector, UG 260/U

Cable, RG 71/U, 93 ohms 6 feet

(7) 216 Mc Output Cable Assembly (W6) - The 216 Mc Output Cable Assembly (W6) shall consist of the following:

Connector, UG 260/U

Connector, UG 261/U

Connector, RG 71/U, 93 ohm, 6 feet

(8) Doppler Input Cable Assembly (W7) - The Doppler Input Cable Assembly (W7) shall consist of the following:

Connector, UG 1033/U

Connector, UG 1033/U

Cable, RG 122/U, 50 ohm, 3 feet

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(9) X-Band Output Flexguide Assembly - The X-Band Output Waveguide Assembly shall consist of the following:

Flange, UG 136/BU

Flange, UG 135/BU

Waveguide, Flexible, length 6 feet

(10) Adapter, Waveguide-to-N Female

(11) Adapter, N Male-to-N Male

(12) Termination, BNC, 100 ohm, ½ watt

(13) Attenuator, 10 db - The Attenuator, 10 db shall have the following electrical characteristics:

Input Connector, TSM Female

Output Connector, BNC Female

Input Impedance, 100 ohm Nom.

Output Impedance, 100 ohm Nom.

Attenuation, 10 db ±1 db

(14) Wrench Set For All Set Screws

(15) Operational Instructions - The following encapsualted drawings shall be supplied with the Test Set.

Parts List and Photograph

Operating Instructions

Schematics

3.4.1.3 Front Panel Controls

(a) Power, On-Off (Miniature circuit breaker type, 3 pole)- Applies power to the test set. Protects all 3 phases of primary power.

(b) Sync Pulse Polarity (Toggle Switch) - Allows test set circuit to be set for incoming trigger polarity.

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(c) Doppler Select, Test-Ext-Zero (Toggle Switch)- Turns on internal doppler generator or allows external input to activate circuit, Zero position allows both X-Band pulses available at output with no doppler modulation.

(d) Meter Select Switch (Rotary Switch, 10 Position)

Position 1	Off (Shorts meter terminals)
Position 2	(Circuit Voltage Monitor)
Position 3	(Circuit Voltage Monitor)
Position 4	(Circuit Voltage Monitor)
Position 5	(Circuit Voltage Monitor)
Position 6	(Circuit Voltage Monitor)
Position 7	(X-Band Zero Set)

This position deactivates X-Band power and allows balance of X-Band thermistor bridge.

Position 8 (X-Band Power Level)

Allows adjustment of the level set attenuator for a power level indication on meter.

Position 9 (X-Band Power Balance)

In this position, the meter shall indicate mid-scale when the power balance control is adjusted for Fo and Fo + 216 Mc power balance.

Position 10 (216 Mc Power Level)

In this position, the 216 Mc RF Switch is turned on continuously. This supplies CW to the COHO output connector and allows adjustment of the 216 Mc Power Level Control for a power level indication on meter, when the 216 Mc Power Attenuator is in the 0 dbm position.

(e) X-Band Pulse Width Switch (Rotary, 3 Position)

Position 1	0.4 usec width
Position 2	0.7 used width
Position 3	Variable width (0.7 usec to 20.0 usec)

(f) X-Band Pulse-CW Switch (Toggle, 2 Position) - Allows X-Band CW to be generated for calibration purposes.

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(g) X-Band Freq. - Calibrated dial. Tunes Klystron to any frequency in range. Dial calibrated in 50 Mc increments. Dial accuracy ±50 Mc.

Fine Tune	Allows fine mechanical adjustment of X-Band frequency.
Electronic Tune	Allows fine electrical adjustment of X-Band frequency. Adjustment Range, 10 Mc min.

(h) X-Band Pwr Bal Control - Uncalibarated. Allows 216 Mc power balance in conjunction with front panel meter.

(i) 216 Mc Pwr Level Control - Uncalibrated. Allows 216 Mc power level set in conjunction with front panel meter.

(j) X-Band Power Attenuator - Calibrated. -10 dbm to -100 dbm. Dial Calibrated every 1 db. Accuracy ±2 db min.

(k) 216 Mc Pwr Attenuator - Calibrated. Rotary step attenuator 0 dbm to + 10 dbm in 10 steps. Accuracy ±0.5 db when terminated by 100 ohms.

(l) X-Band Pulse Width, Variable Control - Calibrated. When X-Band pulse width switch is in variable position, controls pulse width from 0.70 used to 20.0 used. Dial calibrated every 1 used. Dial accuracy ±0.5 used from 1 to 20 used.

(m) X-Band Zero Set Control - Uncalibrated. Allows balance of thermistor bridge circuit.

3.4.1.4 Indicators

(a) Pwr On (Red Lens, Press-To-Test) - Illuminates when power, on-off switch is in the "On" position.

(b) Operate Lamp (Green Lens, Press-To-Test) - Illuminates when 216 Mc crystal oven is at operating temperature and 60 second time delay relay has activated.

(c) COHO Freq Check Internal - (Green Lens, Press-To-Test) Illuminates when COHO input pulse is within 100 Kc of the 216 Mc reference oscillator.

(d) Monitor Meter - Multifunction microammeter.

3.4.1.5 Electrical Connections - Connections to external circuits shall be provided as follows:

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Reference Designation	Receptacle Type	Function
J1	MS3107A-20-4P	Pwr Input
J2	UG-910B/U	Delayed Trig Input
J3	UG-910B/U	Sync Input
J4	UG-910B/U	COHO Freq. Chk External
J5	UG-910B/U	COHO Input
J6	UG-910B/U	216 Mc Output
J7	UG-910B/U	Doppler Input
J8	UG-910B/U	X-Band Δ Freq. Output
None	UG-136B/U	X-Band Output

3.4.1.6 X-Band Output

(a) Frequency Range: 8.5 to 9.6 Gc. Single knob tuning, direct reading dial with accuracy of ±50 Mc.

(b) Frequency Stability: Temperature stability shall be 0.006%/°C minimum. Incidental FM shall be 50 Kc maximum. Line voltage variations of ±10% shall cause a maximum of 0.02% frequency change.

(c) Output Power: -10 to -100 dbm peak pulse power for each output pulse Fo and Fo + 216 Mc. Direct reading dial with accuracy of ± 2 db minimum for Fo output. For calibration purposes, a CW signal shall be available at the waveguide output of the test set.

(d) Spurious Outputs Below 12.4 Gc: Spurious doppler outputs shall be 30 db minimum below desired output. Zero doppler shall be 27 db below desired output. All other spurious outputs shall be 57 db minimum below desired output.

(e) Pulse On-Off Ratio: 80 db minimum

(f) Differential Pulse Amplitude: A circuit shall be provided to allow for adjustment of the differential amplitude between the two X-Band pulses. The circuit shall have sufficient resolution to allow adjustment of the differential amplitude to within 1 db.

(g) Source Impedance: VSWR 1.5:1 maximum

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(h) Pulse Characteristics:

(1) Duration: Measured from the 50% point on the leading edge to the 50% point on the trailing edge with a properly terminated linear detector.

Fixed Pulse Width: 0.4 usec ± 0.05 usec and
0.7 usec ± 0.05 usec

Variable Pulse Width: 0.70 usec to 20.0 usec min.

Pulse Width Jitter: Measured at the 50% point shall be 15 nanoseconds maximum for fixed, 50 nanoseconds maximum for variable.

(2) Rise Time: Measured from 10% to 90% points on leading edge with a properly terminated linear detector; 0.05 usec maximum.

(3) Decay Time: Measured from 10% to 90% points on trailing edge, with a properly terminated linear detector; 0.05 usec maximum.

(4) Pulse Delay: The pulse delay, as measured from the 50% point of the delayed trigger pulse to the 50% point of the X-Band pulse with a properly terminated linear detector, shall not exceed 200 nsec.

(i) High Power Protection: Output isolator protected from accidental application of up to 10 Kw peak, 10 watts average.

3.4.1.7 <u>215 Mc Output</u>

(a) Frequency: 216 Mc ±.001 Mc

(b) Frequency Stability

(1) Long term stability shall be 1 part in 10^6 min.

(2) Short term stability shall be 1 part in 10^8 for 0.5 ms min.

(c) Output Power

(1) 0 dbm to + dbm peak pulse power in 1 db steps.

(2) Accuracy shall be ±0.5 db when terminated by a 100 ohm load impedance.

(3) A 216 Mc CW output shall be generated only when the Meter Select Switch is in the 216 Power Level position.

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(d) Spurious Outputs: 40 db minimum below the desired output.

(e) Pulse On-Off Ratio: 70 db minimum

(f) Source Impedance: VSWR shall be 3.5:1 maximum when diode switch is in "Off" mode and step attenuator is in the +10 dbm position.

(g) Pulse Characteristics

(1) Duration: 0.4 usec ±0.05 usec as measured at the 50% points of the leading and trailing edge with a properly terminated linear detector.

(2) Rise Time: 0.05 usec maximum (10% to 90% points)

(3) Decay Time: 0.05 usec maximum (10% to 90% points)

(4) Pulse Delay: The pulse delay as measured from the 50% point of the sync pulse leading edge to the 50% point of the 216 Mc pulse with a properly terminated linear detector shall not exceed 200 nsec.

(5) Pulse Width Jitter: Measured at the 50% point shall be 15 nanoseconds maximum.

3.4.1.8 COHO Frequency Check

(a) The 216 Mc COHO pulse shall be supplied to the test set, at a 1 mw level into a 100 ohm load with a 1.5:1 VSWR maximum.

(b) The COHO pulse shall be mixed with the output of CW 216 Mc reference oscillator internally within the test set.

(c) The mixer output shall be available for viewing, at a 10 mv minimum level, on an oscilloscope.

(d) A self-test lamp in the test set shall remain illuminated when the COHO input pulse at a PRF of 2400 and a duration of 300 to 400 usec is within 100 Kc of the reference oscillator.

3.4.1.9 X-Band Delta Frequency

(a) The two X-Band frequencies generated by the test set shall be mixed, internally, for all positions of the Meter Selector switch except the X-Band Power Calibrate, Zero Set -12 VDC, and -565 VDC positions.

(b) The X-Band Delta Frequency shall be available at the test set front panel.

(c) The level of the X-Band Delta Frequency shall be -23 dbm nominal.

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3.4.1.10 Recovered Doppler Frequency

(a) The X-Band Delta Frequency may be connected, by means of an external jumper cable, to the COHO frequency input connector at the test set front panel.

(b) The recovered doppler signal shall be at a minimum level of 10 mv. This output will be available at the COHO frequency check connector at the test set front panel.

3.4.1.11 Doppler Input Frequency

(a) This input shall cause the 216 Mc frequency difference of the X-Band pulse outputs to increase with doppler modulation.

(b) Characteristic: Sinusoidal

(c) Frequency: 20 to 2000 cps. Rate of change of frequency is 70 cycles/sec/sec maximum.

(d) Amplitude: 16 volts peak-to-peak ± 1 volt into a 1000 ohm load. Amplitude shall be constant to within ± 0.5 volts over the frequency range of 20 to 2000 cps.

3.4.1.12 Internal Doppler Generator

(a) An internal doppler frequency shall be supplied to allow normal test set operation in the absence of an external doppler input. The frequency shall be 10 kc + 1 kc.

(b) A zero doppler frequency shall be supplied when the Doppler Select switch is in the zero position.

3.4.1.13 External Sync Pulse Characteristics

(a) Duration: (Measured at 50% points) 0.4 to 5.0 usec

(b) Rise Time: (Measured at 10% to 90% points) 0.1 usec to 1.0 usec

(c) Amplitude: 5 volts peak to 25 volts peak

(d) Polarity: Positive or negative, switch selectable

(e) PRF: 40 to 4000 pps

(f) Input Impedance: 1000 ohms nominal

3.4.1.14 External Delayed Trigger Pulse Characteristics

(a) Duration: (Measured at 50% points) 0.4 to 5.0 usec

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(b) Rise Time: (Measured at 10% to 90% points) 0.1 usec to 1.0 usec

(c) Amplitude: 5 volts peak to 25 volts peak

(d) Polarity: Positive or negative, switch selectable

(e) PRF: 40 to 4000 pps

(f) Input Impedance: 1000 ohms nominal

3.4.1.14 External Delayed Trigger Pulse Characteristics

(a) Duration: (Measured at 50% points) 0.4 to 5.0 usec

(b) Rise Time: (Measured at 10% to 90% points) 0.1 usec to 1.0 usec

(c) Amplitude: 5 volts peak to 25 volts peak

(d) Polarity: Positive

(e) PRF: 40 to 4000 pps

(f) Input Impedance: 1000 ohms nominal

3.4.2 Case, Test Set, CY-6615/APM-299 - The Test Set, Radar shall be provided with a combination case which shall conform to Specification MIL-T-21200.

3.4.2.1 Size - The maximum dimensions of Case, Test Set, CY-6615/APM-299 shall be 23 inches by 22 inches by 18 inches.

3.4.3 Cover, Test Set, CW-1048/APM-299 - The Cover, Test Set, CW-1048/APM-299 shall provide stowage for the accessory items listed in 3.4.1.3(p).

3.4.3.1 Size - The maximum dimensions of Cover, Test Set, CW-1048/APM-299 shall be 23 inches by 10.25 inches by 18 inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(a) Preproduction Tests

(b) Initial Production Tests

(c) Acceptance Tests

(d) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on a Test Set representative of the production Test Sets to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted, so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts, giving their specified voltage and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the Test Set meets all the requirements of this specification and the contract. Preproduction test shall include environmental tests in accordance with procedures of MIL-T-21200 and interference tests in accordance with MIL-I-6181. Unless specified by contract action, fungus, sand, dust, explosion and crash safety tests will not be required.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each Test Set shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for rejection of Test Sets already accepted by the Government; it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production Test Sets shall be selected and sent, at the contractor's expense, to a designated Government laboratory for tests. This Test Set shall be selected by the procuring activity after it has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - The Test Set may be subjected to any and all tests the procuring activity deems necessary to assure that the production Test Set is equivalent in design, construction, workmanship, performance, and quality to the previously approved preproduction sample and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete Test Set submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the test sample shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the Test Set for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Reliability Assurance Tests

(d) Special Tests

4.4.1 Individual Tests - Each Test Set submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product

(b) Operational Test

(c) Manufacturing Run-in Test

4.4.1.1 Examination of Product - Each Test Set shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each Test Set shall be operated long enough to permit its temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory Test Set operation.

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4.4.1.3 Manufacturing Run-In Test - Each Test Set shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the Test Set. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the Test Set.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The Test Set shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the Test Set shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the Test Set in 2 directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the Test Set shall be mechanically cycled, periodically, through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate the cause of failure from future Test Sets. A record shall be kept of all failures. The 6-hours period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Test Sets selected for sampling tests shall first have passed the individual tests. Test Sets shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling test shall be subjected to the following tests:

(a) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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(b) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(c) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests are not required unless specified by the contract.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of Test Sets for special tests shall be made as follows:

(a) On an early Test Set after an engineering or material change.

(b) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a test in the individual test to check all Test Sets for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on Test Sets that have passed the individual tests. The life test shall be performed under the conditions specified in

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4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Test Sets which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of Test Sets equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient ($30^{o}C \pm 10^{o}C$)
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room Ambient
Input Voltage	115 ±10% volts AC (at lowest applicable frequency)
	27.0 ±2.0 volts DC
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the Test Set shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the Test Set, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

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(a) Part type number

(b) The circuit reference symbol number

(c) The part function

(d) Name of the manufacturer

(e) Nature of the failure

(f) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - A Test Set which has been subjected to the life test shall be reconditioned as follows:

(a) On completion of the life test, the Test Set shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the Test Set for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Test Sets which have been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part, or complete Test Set shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.9 Rejection and Retest - Test Sets which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original inspection shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - Alignment and test of Radar System

6.2 Performance Characteristics - Nominal performance characteristics shall be determined at input voltages of 115 V ±1.0 VAC 3 phase, 400 cycle. (See 3.3.8 for Standard Conditions.) Limiting performance characteristics shall be determined at the highest and lowest values of specified input voltages.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accomplished by complete supporting information.

6.4 As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supply), the non-repairable subassembly should evidence a meantime-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - the contract shall have precedence over any specification.

(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.6 The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Selection of applicable levels of packaging and packing (See 5.1)

6.8 Notation of Revisions - A bar in the margin denotes a change made in the adjacent paragraph, from the previous version of the document. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

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Naval Air Systems Command
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81761(AS) TEST SET, RADAR AN/APM-299

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO. IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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